Scope of Work (SoW): Tokenization & Digital Asset Compliance Platform (Similar to Tokeny.com)

1. Project Overview

Virtual Height will design, develop, and deploy an **enterprise-grade Tokenization Platform** enabling organizations to issue, manage, and transfer **regulated digital assets** (security tokens, real-world asset tokens, utility tokens).
The platform will incorporate **compliance automation**, **investor onboarding**, **KYC/AML workflows**, and **token lifecycle management** using **blockchain infrastructure**.

2. Core Objectives

- Enable organizations to **tokenize real-world assets** (equity, real estate, bonds, funds, commodities).
- Provide **end-to-end compliance** (KYC/AML, whitelisting, transfer restrictions).
- Offer **modular issuance tools** to launch tokens quickly.
- Deliver **secure and scalable smart contracts** and APIs.
- Provide an **admin dashboard** for issuers and **investor portal** for users.
- Ensure interoperability with **multiple blockchains (Ethereum, Polygon, Hyperledger, Bharat Blockchain)**.

3. Major Modules & Deliverables

3.1 Token Issuance Engine

- Create a no-code or low-code token creation interface.
- Configure:
    - Token name, supply, decimals
    - Asset type: equity / real estate / funds / debt
    - Compliance rules (jurisdictions, lock-in periods, transfer rights)
- Deploy ERC-20 / ERC-3643 (T-REX) / ERC-721 / custom token standards.
- Automated contract deployment using secure DevOps pipelines.

Deliverables:

- Token issuance UI
- Smart contract templates
- Compliance rule engine

3.2 Compliance & Identity Management (KYC/AML)

- Integrated workflows for:
    - KYC onboarding
    - AML checks
    - Accreditation verification
    - Document uploads & verification
- Automated **whitelisting** & status management.
- Support for global compliance frameworks:

    - SEBI
    - FATF
    - EU AML directives

Deliverables:

- Investor KYC portal
- Issuer compliance dashboard
- Automated verification API integrations

3.3 Investor & Issuer Portals

Investor Portal

- User registration, 2FA, wallet connection
- Portfolio dashboard
- Token purchase & redemption (fiat/crypto)
- Secure document vault

Issuer Portal

- Dashboard to monitor investors, token issuance, compliance
- Cap table management
- Transfer approvals
- Corporate actions (dividends, voting, announcements)

Deliverables:

- Web-based portals for investors & issuers
- Role-based access control

3.4 Token Lifecycle Management

- Automated modules for:
    - Primary issuance
    - Secondary transfers
    - Lock/unlock periods
    - Vesting schedules
    - Corporate voting
    - Redemption or buy-back

Deliverables:

- Transaction management system
- Event-trigger smart contracts

3.5 Marketplace / Secondary Trading (Optional)

- Compliant peer-to-peer token transfers
- Integration with regulated ATS/exchanges
- Order matching engine (optional)
- Built-in OTC deal module

Deliverables:

- P2P trading interface
- Trade compliance engine

3.6 Blockchain Infrastructure

- Support for multiple networks:
    - Ethereum mainnet/testnets
    - Polygon
    - Bharat Blockchain Network
    - Private/permissioned blockchain for government use
- Node management & monitoring
- Gas abstraction and meta-transactions (optional)

Deliverables:

- Scalable blockchain node infrastructure
- Interoperable deployment scripts

3.7 APIs & Integrations

- REST & Web3 APIs for:
    - Token issuance
    - Wallet management
    - Investor onboarding

    - Compliance checks
- Integration with 3rd party:
    - KYC/AML providers
    - Payment gateways (fiat & crypto)
    - Custodians (if required)

Deliverables:

- API documentation
- Postman collection

3.8 Security & Audits

- Smart contract audits
- Penetration testing
- OWASP compliance
- Role-based authentication & logging
- Secure cloud architecture

Deliverables:

- Audit reports
- Security hardening documentation

3.9 Admin Master Console

- Platform-level administration
- User management
- Issuer onboarding & approvals
- System logs & audit trails
- Analytics dashboards

Deliverables:

- Super Admin panel
- Dynamic configuration controls

4. Technology Stack

Frontend

- React / Next.js
- Tailwind / Material UI
- Wallet integrations: WalletConnect, Metamask, custom custodial wallets

Backend

- Node.js / NestJS
- Microservices architecture
- PostgreSQL / MongoDB
- Redis for caching

Blockchain

- Solidity
- Hardhat / Foundry
- ERC-20 / ERC-3643 / ERC-1400 standards

Infra

- AWS / Azure
- Kubernetes / Docker
- CI/CD pipelines

5. Project Timeline (Sample 16–20 Weeks)

Phase	Duration	Deliverables
Discovery & Architecture	2 weeks	BRD, wireframes, architecture
UI/UX Design	2 weeks	Design system, UX flows
Backend + Smart Contracts	6 weeks	APIs, token engine, compliance engine
Frontend Development	6 weeks	Portals, dashboards
Integrations	2 weeks	KYC, payments, blockchain
Testing & Security Audits	2 weeks	QA, UAT, audits
Deployment & Handover	1 week	Production release

6. Post-Launch Support

- 3-month warranty period
- SLA-driven AMC
- Optional DevOps & blockchain maintenance
- Feature enhancements

7. Intellectual Property

- All custom development, smart contracts, and source code will belong to the client.
- Open-source components will follow respective licenses.